Year 10 Mathematics

Sets 1

9.4 Practical Problems

Practical Problems

STARTER · 5 MIN

Date: _____ **Title:** Practical Problems

LEARNING OBJECTIVES

- Use Venn diagrams to solve real-life problems
- Complete Venn diagrams from given information
- Use algebra to find unknown values in Venn diagrams

STARTER *Answer in your book. No calculators.*

1. Work out 30 − 12
2. Work out 25 − 8 − 6
3. Solve $x + 12 = 30$
4. Solve $40 - x + x + 15 - x = 48$

Starter — Answers

STARTER

Tick or correct in green pen

1. $30 - 12 = \mathbf{18}$
2. $25 - 8 - 6 = \mathbf{11}$
3. $x = 30 - 12 = \mathbf{18}$
4. $55 - x = 48, \quad x = \mathbf{7}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use Venn diagrams to solve real-life problems	☐
SILVER	I can complete Venn diagrams from given information	☐
GOLD	I can use algebra to find unknown values in Venn diagrams	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Practical Problems

INTRODUCTION · 3 MIN

Use a Venn diagram to organise information about groups of people or objects.

Start in the middle

Put the number in both groups first.

Like football and tennis: 7

Subtract

Find 'only' regions by subtracting.

25 like football: 25 − 7 = 18

Outside

Total minus everything inside.

40 − 35 = 5

Unknowns

Use x for an unknown region, then solve.

Add all regions = total

Examples 1 and 2 — Completing a Venn Diagram

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

40 students. 25 like football (F), 17 like tennis (T),

7 like both. Complete the Venn diagram.

Example 2

Use the diagram to find how many students like football or tennis.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Both = 7

Only F = 25 − 7

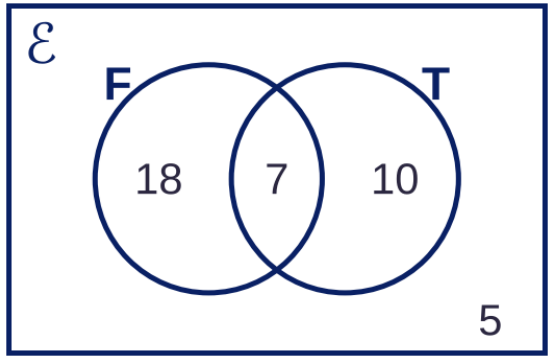
= 18

Only T = 17 − 7

= 10

Neither = 40 − 35

= 5



Tip: always start with the overlap.

Example 2

18 + 7 + 10

= 35 students

Tip: 'or' means the union.

Task 1

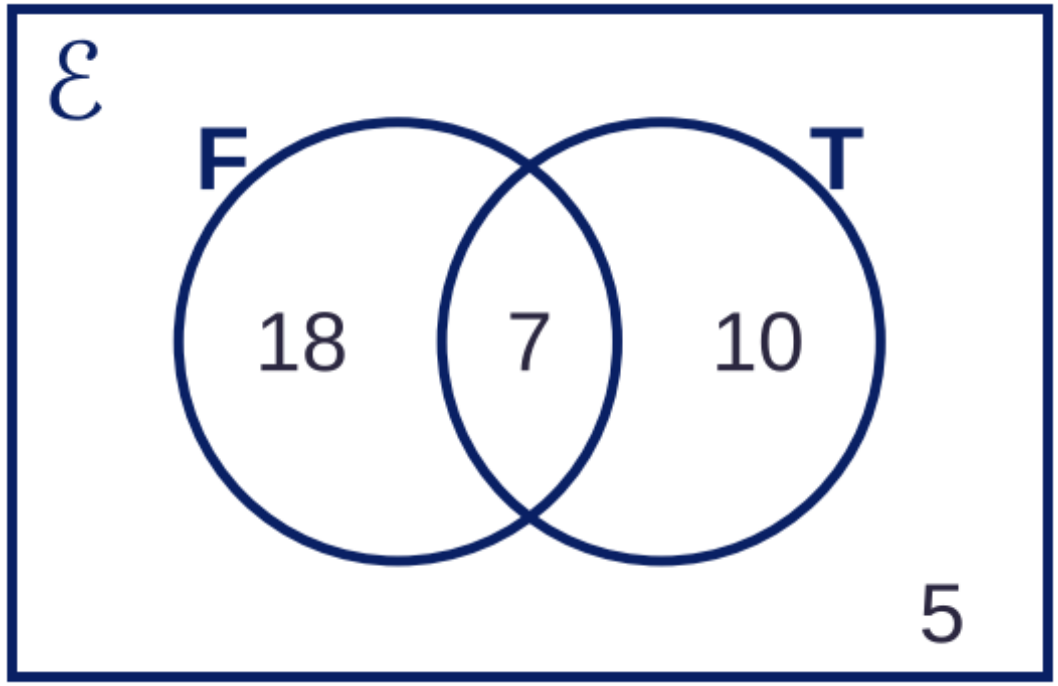
PRACTICE · 9 MIN

BRONZE

Use the Venn diagram of 40 students

1. How many like football only?
2. How many like tennis?
3. How many like neither?
4. How many like exactly one sport?
5. What fraction of the students like both?

★ How many do not like tennis?



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. Football only = **18**

2. 10 + 7 = **17**

3. Neither = **5**

4. 18 + 10 = **28**

5. $\frac{7}{40}$

★ 18 + 5 = **23**

Examples 3 and 4 — Real-Life Problems

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Example 3

50 people. 30 have a cat, 22 have a dog, 9 have both.

How many have neither?

Example 4

32 students study French or Spanish or both.

20 study French and 18 study Spanish.

How many study both?

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$\begin{aligned} \text{Only cat} &= 30 - 9 \\ &= 21 \\ \text{Only dog} &= 22 - 9 \\ &= 13 \\ 21 + 9 + 13 &= 43 \\ \text{Neither} &= 50 - 43 \\ &= \mathbf{7} \end{aligned}$$

Tip: draw the diagram first.

Example 4

$$\begin{aligned} 20 + 18 &= 38 \\ \text{Both} &= 38 - 32 \\ &= \mathbf{6} \end{aligned}$$

Tip: the overlap was counted twice.

Task 2

PRACTICE · 9 MIN

SILVER

Draw a Venn diagram for each question

Hint $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

1. 60 students: 35 play piano, 28 play guitar, 12 play both. How many play neither?
2. 45 people: 26 drink tea, 21 drink coffee, 8 drink both. How many drink only tea?
3. 30 students all take art or music. 18 take art, 20 take music. How many take both?
4. 100 shoppers: 64 bought bread, 48 bought milk, 20 bought both. How many bought neither?
5. In a class of 28, 15 have a sister, 12 have a brother, 6 have both. How many have neither?

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $23 + 12 + 16 = 51, \quad 60 - 51 = \mathbf{9}$

2. $26 - 8 = \mathbf{18}$

3. $18 + 20 - 30 = \mathbf{8}$

4. $44 + 20 + 28 = 92, \quad 100 - 92 = \mathbf{8}$

5. $9 + 6 + 6 = 21, \quad 28 - 21 = \mathbf{7}$

Examples 5 and 6 — Using Algebra

EXAMPLES · 5 MIN

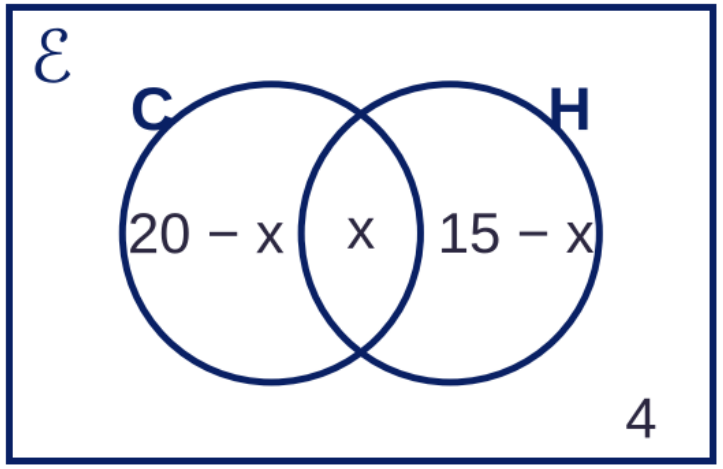
GOLD

Exam-style: show all your working

Example 5

(3 marks)

35 people: 20 like coffee (C), 15 like hot chocolate (H), 4 like neither. Find x.



Example 6

(2 marks)

Using x = 4, how many people

like coffee only?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$20 - x + x + 15 - x + 4 = 35$$

$$39 - x = 35$$

$$\mathbf{x = 4}$$

Tip: add all the regions and make them equal the total.

Example 6

$$20 - x = 20 - 4$$

Coffee only **= 16**

Tip: substitute x back in.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. 50 students: 30 like maths, 26 like science, 6 like neither. How many like both? (3 marks)

2. 40 people: 22 swim, 25 run, x do both, 3 do neither. Find x. (3 marks)

3. In a group, 2x like A only, x like both, 10 like B only, 4 like neither. The group has 38 people. Find x. (3 marks)

4. Using your answer to question 3, how many like A? (1 mark)

5. 60 people: 35 read news, 30 read sport, 2x read both, x read neither. Find x. (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $30 + 26 - x + 6 = 50, \quad x = \mathbf{12}$
2. $22 + 25 - x + 3 = 40, \quad x = \mathbf{10}$
3. $3x + 14 = 38, \quad x = \mathbf{8}$
4. $2x + x = \mathbf{24}$
5. $65 - 2x + x = 60, \quad x = \mathbf{5}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE 30 students: 18 like rice, 15 like pasta, 6 like both. How many like neither?

SILVER In the same group, how many like exactly one food?

GOLD 40 people: 24 own a car, 20 own a bike, x own both, 4 own neither. Find x.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $12 + 6 + 9 = 27, \quad 30 - 27 = \mathbf{3}$

SILVER $12 + 9 = \mathbf{21}$

GOLD $24 + 20 - x + 4 = 40, \quad x = \mathbf{8}$

Exam Practice — Completing a Venn Diagram

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

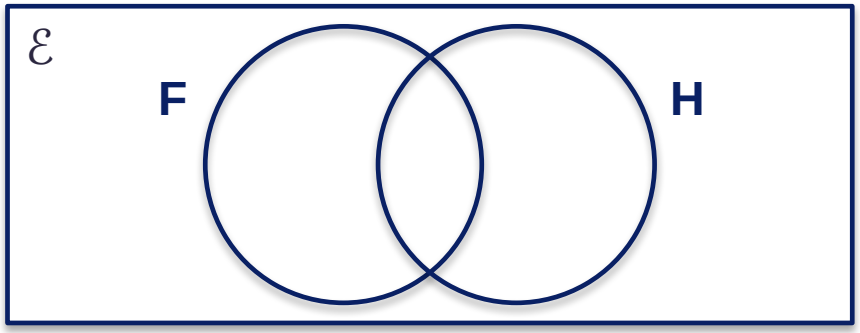
(2 marks)

30 students were asked about sport.

18 play football (F).

14 play hockey (H). 7 play both.

Complete the Venn diagram.



Question 2

(2 marks)

Use your diagram from Question 1.

(a) How many students play exactly one sport? (1)

.....

(b) What fraction of the students play neither sport? (1)

.....

Exam Practice — Real-Life Problems

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Question 3

(3 marks)

There are 60 people in a group.

35 have a cat. 28 have a dog.

12 have both a cat and a dog.

How many have neither?

.....

Question 4

(3 marks)

45 students all study French or Spanish or both.

30 study French. 24 study Spanish.

How many study only Spanish?

.....

Exam Practice — Using Algebra

EXAM PRACTICE · 10 MIN

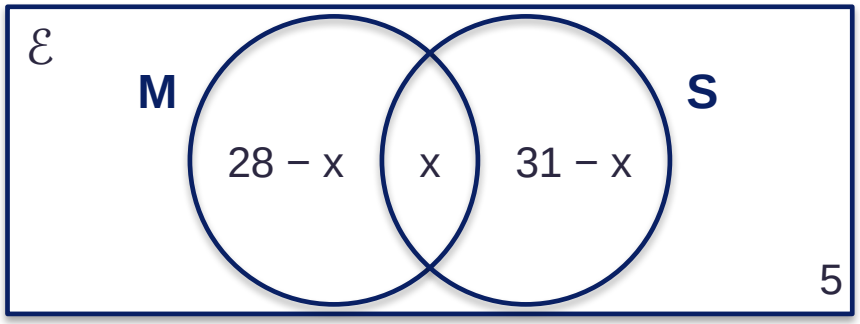
GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(4 marks)

50 students were asked if they like maths (M) or science (S).



(a) Find x. (3)

x = ...

(b) How many like maths only? (1)

.....

Question 6

(3 marks)

A club has 48 members.

3x play only chess.

x play chess and draughts.

2x play only draughts.

6 play neither.

How many members play chess?

.....

Exam Practice — Mark Scheme

CORRECTIONS

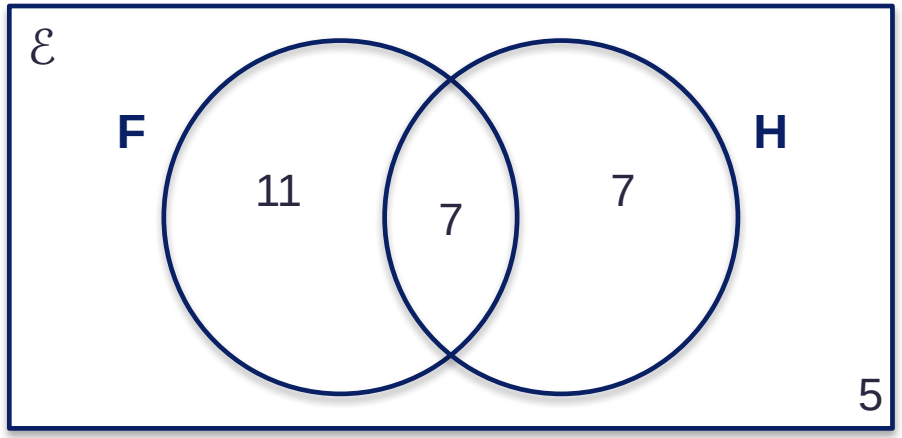
BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$18 - 7 = 11,\ \ 14 - 7 = 7$ **B1**

7 in middle, 5 outside **B1**



Tip: always start with the overlap.

Question 2

(a) 11 + 7 = 18 **B1**

(b) $\frac{5}{30} = \frac{1}{6}$ **B1**

Tip: neither = 30 − 25 = 5 students.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$35 - 12 = 23,\ \ 28 - 12 = 16$	M1
$23 + 12 + 16 = 51$	M1
$60 - 51 = 9$	A1

Tip: draw the Venn diagram first.

Question 4

$30 + 24 - 45 = 9$ study both	M1
$24 - 9$	M1
$= 15$	A1

Tip: the overlap was counted twice.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$28 - x + x + 31 - x + 5 = 50$ — M1

$64 - x = 50$ — M1

(a) $x = 14$ — A1

(b) $28 - 14 = 14$ — B1

Tip: add all regions, set equal to the total.

Question 6

$3x + x + 2x + 6 = 48$ — M1

$6x = 42,\ x = 7$ — A1

$3x + x = 28$ — A1

Tip: chess = only chess + both.